

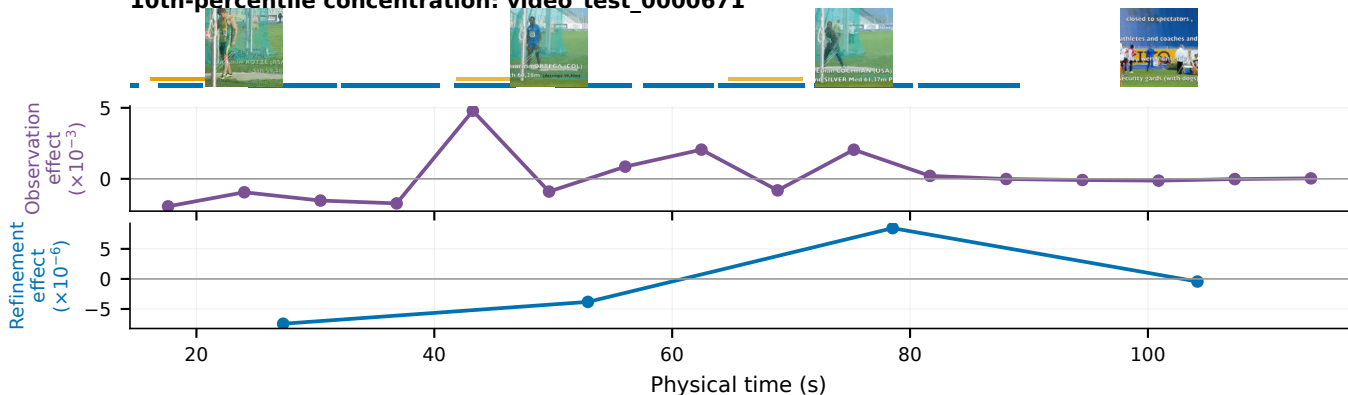
Fixed RGB
& physical time

Frozen dense
TAD model

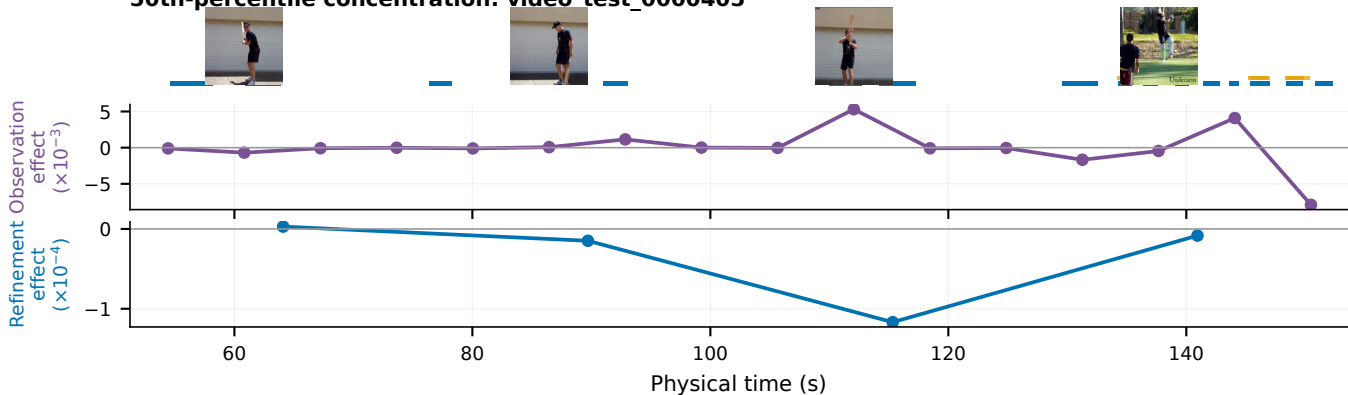
Original prediction
+ cls / loc loss

Measure task-loss effects of local observation and computation changes.

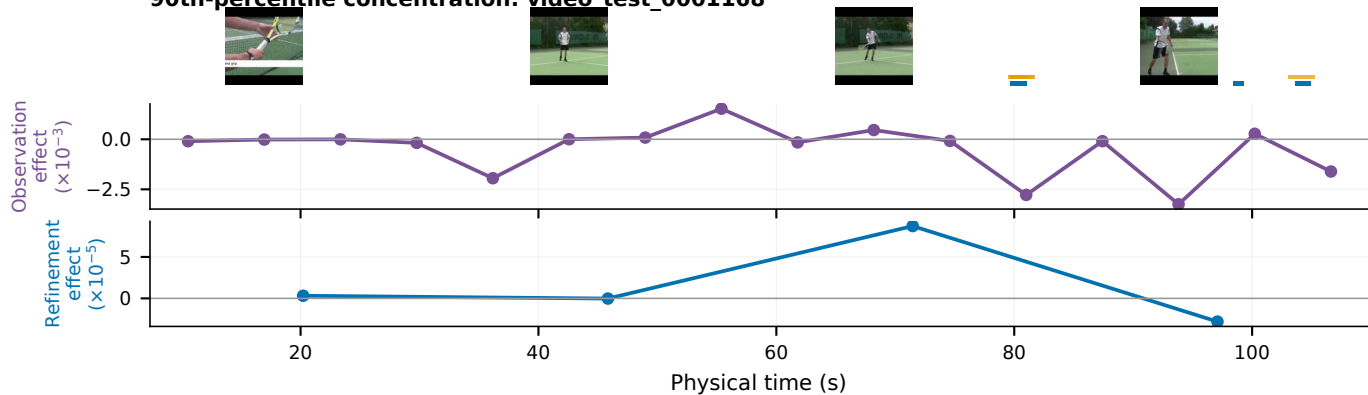
10th-percentile concentration: video test_0000671

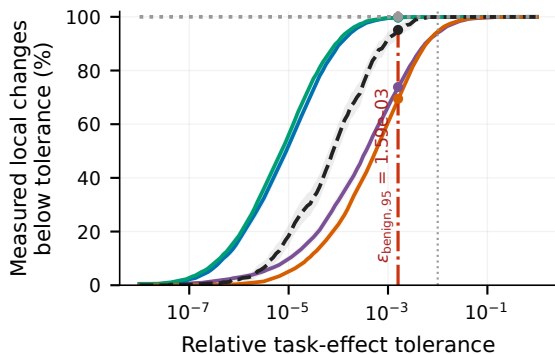


50th-percentile concentration: video test_0000405



90th-percentile concentration: video test_0001168

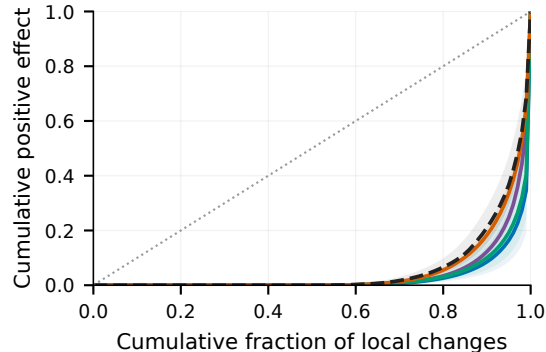
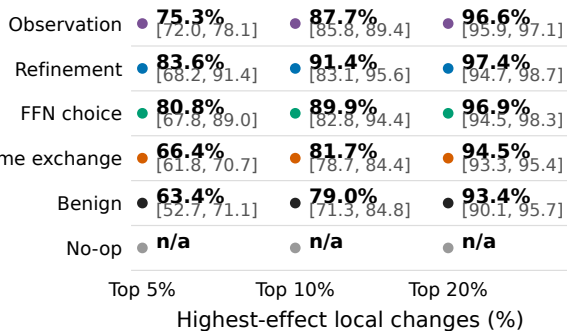
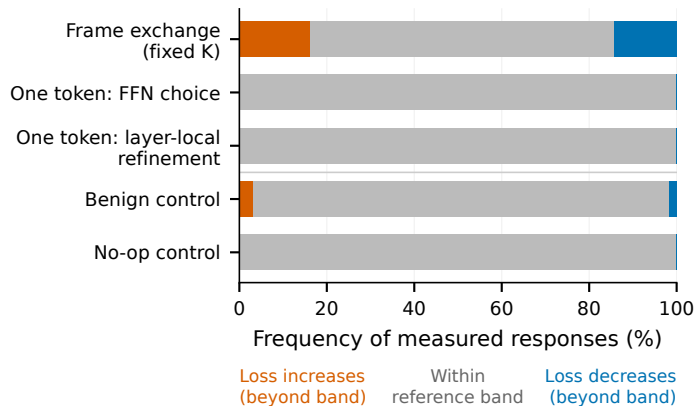


(a) AdaTAD-S: effect magnitude

Above benign reference (%):

Obs 26.17 D 0.21 S 0.26

T 30.53 Benign 4.88 No-op 0.00

(b) Positive-part concentration**(c) Positive measured effect (%)****(d) One local choice change**

— Observation replacement

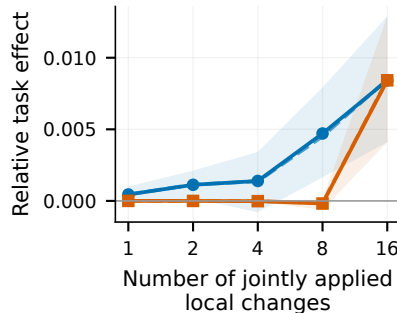
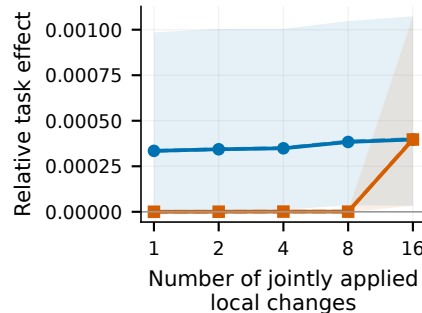
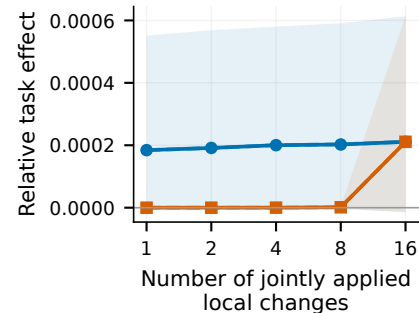
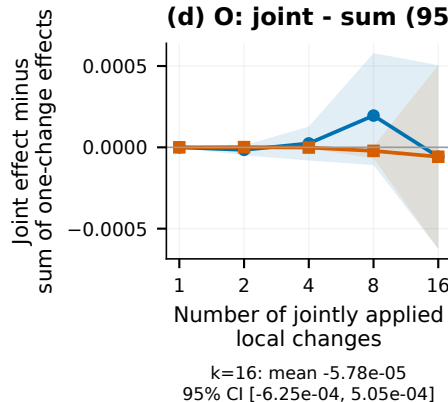
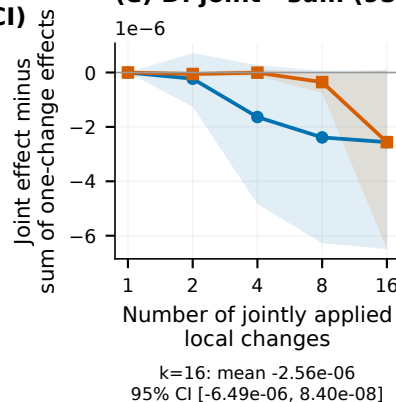
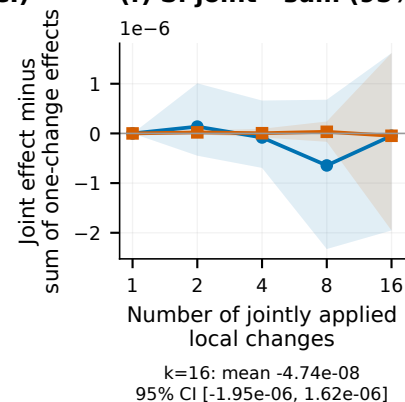
— Layer-local refinement

— Spatial FFN choice

— Frame exchange (eligible only)

— Benign control

... No-op control

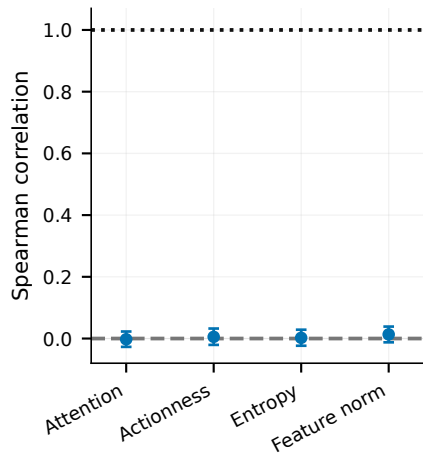
(a) S model / O: joint vs sum**(b) S model / D: joint vs sum****(c) S model / S: joint vs sum****(d) O: joint - sum (95% CI)****(e) D: joint - sum (95% CI)****(f) S: joint - sum (95% CI)**

● Joint effect: stratified local changes

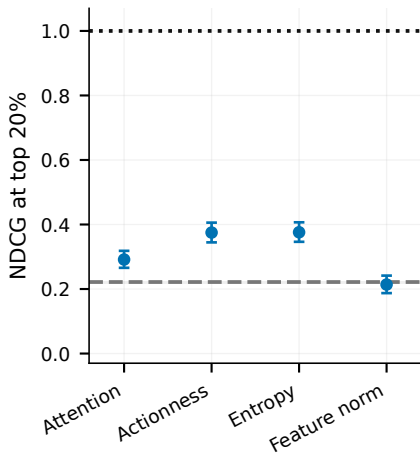
— Sum of one-change effects

■ Joint effect: changes with small one-change effect magnitudes

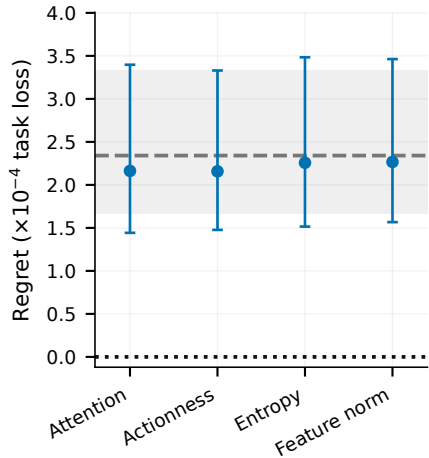
(a) AdaTAD-S / D choices
Rank correlation



(b) AdaTAD-S / D choices
Top-20% relevance



(c) AdaTAD-S / D choices
Signed-value regret



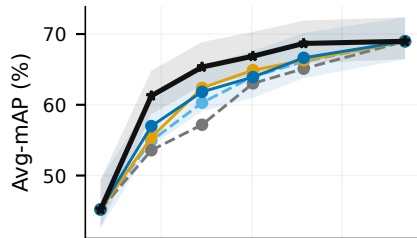
-- Random (exact expectation)

.... Actual-value ranking

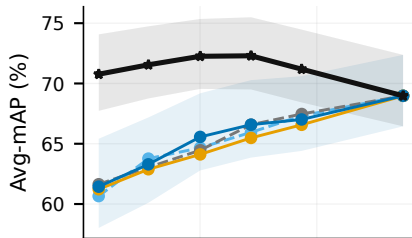
Proxy

AdaTAD-S: compute allocation

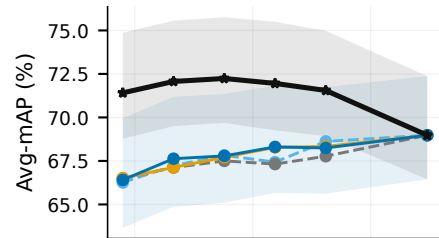
(a) T: Frame selection



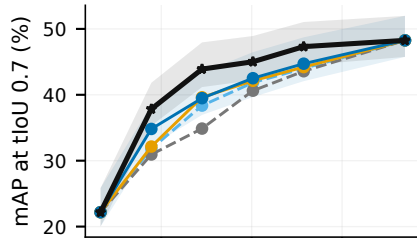
(b) D: Layer-local refinement



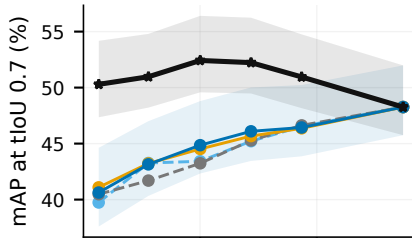
(c) S: FFN allocation



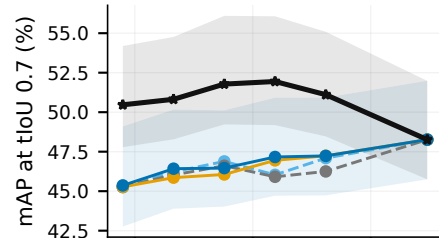
(d) T: Frame selection



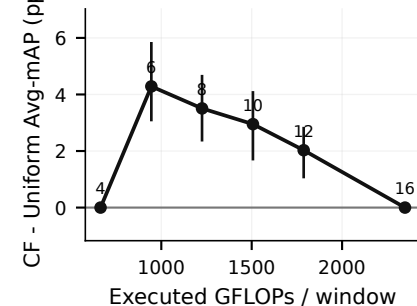
(e) D: Layer-local refinement



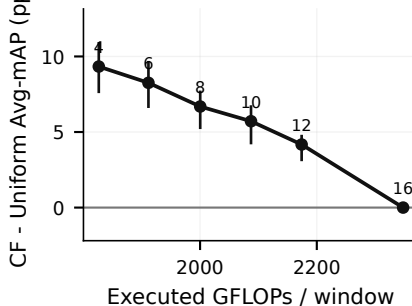
(f) S: FFN allocation



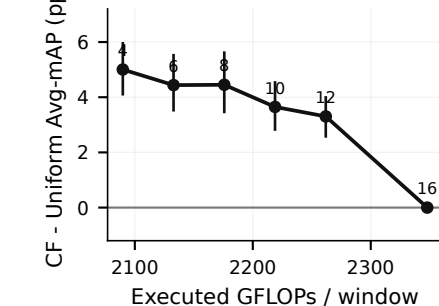
(g) T: paired difference



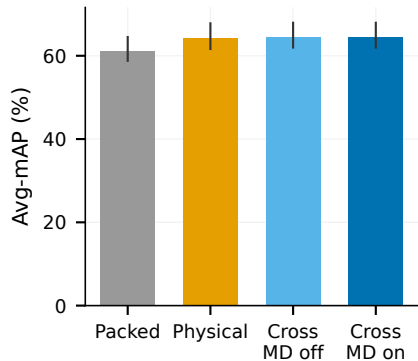
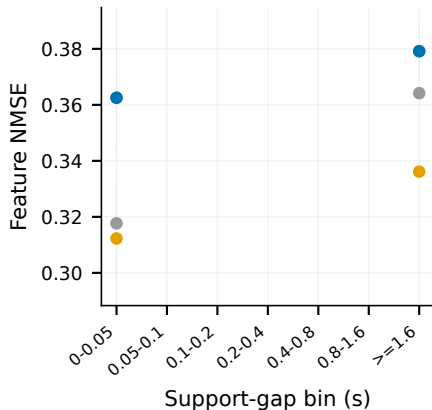
(h) D: paired difference



(i) S: paired difference



—●— Random
 —●— Attention (diagnostic)
 —●— Marginal CF reference
—●— Static (development)
 —●— Uniform

(a) V2-S: fixed-support AP**(b) Fidelity across support gaps****(c) Matched endpoint error**